

TANVIR HASAN RATUL

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

OBJECTIVE

Computer Science undergraduate seeking a Software Engineering / Backend Developer internship. Experienced building FastAPI backends, relational-database-driven applications, and Python data tools; comfortable with REST API design, SQL, and object-oriented programming.

EDUCATION

Ahsanullah University of Science and Technology (AUST)

B.Sc. in Computer Science and Engineering (CSE) — 3rd Year, 2nd Semester | CGPA: 3.223 / 4.00 (in progress)

St. Joseph Higher Secondary School & College

Higher Secondary Certificate (HSC) — GPA: 5.00 / 5.00

Narayanganj Ideal School

Secondary School Certificate (SSC) — GPA: 5.00 / 5.00

TECHNICAL SKILLS

Languages: Python, SQL, C++

Backend & APIs: FastAPI, REST APIs, SQLite, SQL Server (SSMS), pyodbc

Data & Tools: Pandas, Streamlit, Power BI, Power Query, Excel, Git/GitHub

Core CS: Object-Oriented Programming, Data Structures & Algorithms

PROJECTS

Office Watch — Real-Time Office Energy Monitoring System | Python, FastAPI, SQLite, WebSockets, JavaScript

- Built a FastAPI backend acting as the single source of truth for a live web dashboard and a separate Discord bot, both consuming shared device state over REST and WebSocket endpoints
- Designed a 4-table SQLite persistence layer (devices, device events, daily energy usage, alert history) so device state and alert history survive backend restarts
- Implemented after-hours and continuous-on alert detection with a queryable alert-history API endpoint; built for a national-level hackathon (Techathon Nationals & Rover Summit preliminary round)

Internship Application Tracker | Python, SQL Server, pyodbc, OOP

- Designed a relational schema (Student, Company, Application, Interview) in SQL Server and built a Python data-access layer using pyodbc
- Implemented CRUD operations and keyword search (SQL LIKE queries) through an object-oriented Python structure with exception handling, plus a separate reporting module for aggregated application-status summaries

Weather Dashboard | Python, REST API, Pandas, Streamlit

- Built an interactive Streamlit dashboard that fetches live weather and next-day forecasts from the OpenWeatherMap REST API
- Used Pandas to log every search to a CSV-backed history with filtering by city/date, and computed forecast min/max/average temperature with a trend chart

Football Player Performance Dashboard | SQL Server (SSMS), T-SQL, Power Query, Power BI, Excel

- Cleaned and transformed a Kaggle football dataset using T-SQL, then modeled relationships and built a 4-page interactive Power BI dashboard (Top Performers, Assists & Contributions, Young Talents, Future Predictions)
- Applied Power Query transformations and DAX-based calculated fields to support filtering by nation, position, age, and market value

Additional Power BI project: Cristiano Ronaldo Club Career Stats Dashboard (2003–2025) — interactive KPI dashboard analyzing goals, assists, and club-wise performance trends.

LEADERSHIP & ACTIVITIES

- Deputy Coordinator, Department of Publication — AUST Programming and Informatics Club (AUST PIC)
- Member, Josephite Math Club

CERTIFICATIONS

- Google Prompting Essentials Specialization — Coursera (Google Career Certificates)